

Mean-Shift tracking

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I. INTRODUCTION

II. EXPERIMENTS

A. Kalman filtering

In this section we derived the input matrices (see Appendix) for the Random walk (RW), Nearly constant velocity (NCV) and Nearly constant acceleration (NCA) motion models. We tested all these models on three different trajectories.

Firstly on Figures 1, 2 and 3 we can see the trajectories and their filtered counterparts in dependence to different motion models as well as constants for dynamics noise(q) and measurement covariance(r). On all three plots, for all three motion models, we can see that when setting a large q and small r the Filtered measurements follow the measurements nicely, which is expected since we assign more uncertainty to the dynamics, so our tracker mostly uses the measurements for prediction.

On the other hand, when setting a high r and low q we are relying more on the motion model. We can see that because the tracker relies mostly on the motion models, the random walk cannot follow the path fast enough. As for the NCV and NCA models, we can see that the paths start to differ a lot from the measurements, especially in the second plot. This is not necessarily a bad thing, since in situations where our measurements are poor, the tracker will perform better, since it will rely more on the motion model.

The optimal parameters are dependent on the choice of motion model and specific application. For RW we should rely more on the measurements so maybe $q = 5$ and $r = 1$ would be acceptable. For the NCV and NCA models we can rely more on the motion model so we can pick a larger r .

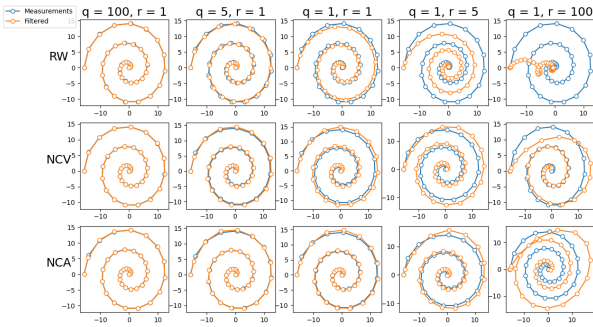


Figure 1. Kalman filtering method on the spiral curve

B. Results of the basic particle filter tracker implementation

Here we implemented the particle filter tracker using the NCV motion model and color histograms for the visual model. We can observe the results for the entire VOT 2014 dataset in Table I. Note that we used an update speed $\alpha = 0.00001$,

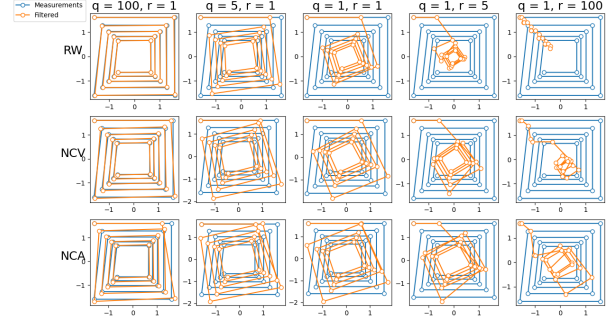


Figure 2. Kalman filtering method on the square spiral curve

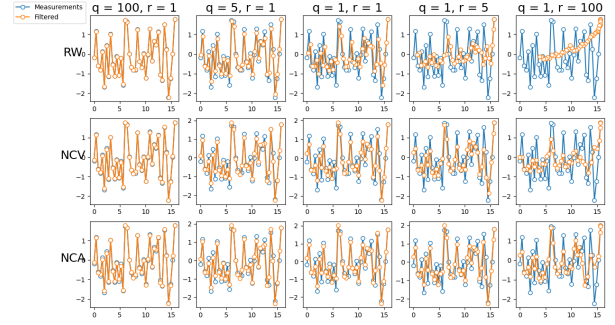


Figure 3. Kalman filtering method on random gaussian noise

kernel size $\sigma = 1$, 16 bins in the histogram, the dynamic noise constant was 80% of the size of the smaller dimension of the patch and the number of particles was set to 100.

Metric	Value
Average Overlap	0.52
Total Failures	32
Average Speed (FPS)	86.76

Table I

BASIC PARTICLE TRACKER PERFORMANCE

C. Testing using different motion models

We tested the tracker using all of the three aforementioned motion models. For each of the models we tested various different dynamic noise constants(q). We can see the number of failures as well as average overlap when using for different parameters in Figure 4.

D. Testing the influence of number of particles

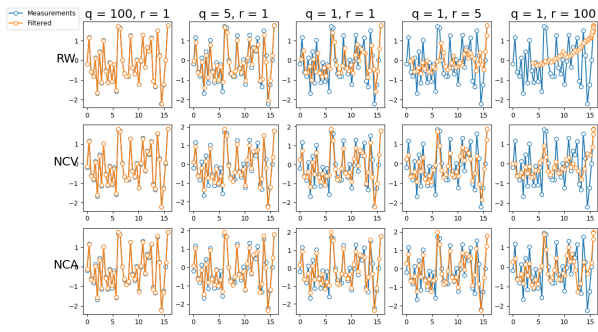


Figure 4.

III. APPENDIX

Here we present the matrices derived for the motion models

A. Random walk

$$\mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \phi = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} \Delta t q & 0 \\ 0 & \Delta t q \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

B. Nearly constant velocity

$$\mathbf{X} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \Phi = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} \frac{\Delta t^3 q}{3} & 0 & \frac{\Delta t^2 q}{2} & 0 \\ 0 & \frac{\Delta t^3 q}{3} & 0 & \frac{\Delta t^2 q}{2} \\ \frac{\Delta t^2 q}{2} & 0 & \Delta t q & 0 \\ 0 & \frac{\Delta t^2 q}{2} & 0 & \Delta t q \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

C. Nearly constant acceleration

$$\mathbf{X} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ \ddot{x} \\ \ddot{y} \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \Phi = \begin{bmatrix} 1 & 0 & \Delta t & 0 & \frac{\Delta t^2}{2} & 0 \\ 0 & 1 & 0 & \Delta t & 0 & \frac{\Delta t^2}{2} \\ 0 & 0 & 1 & 0 & \Delta t & 0 \\ 0 & 0 & 0 & 1 & 0 & \Delta t \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{Q} = \begin{bmatrix} \frac{\Delta t^5 q}{20} & 0 & \frac{\Delta t^4 q}{8} & 0 & \frac{\Delta t^3 q}{6} & 0 \\ 0 & \frac{\Delta t^5 q}{20} & 0 & \frac{\Delta t^4 q}{8} & 0 & \frac{\Delta t^3 q}{6} \\ \frac{\Delta t^4 q}{8} & 0 & \frac{\Delta t^3 q}{3} & 0 & \frac{\Delta t^2 q}{2} & 0 \\ 0 & \frac{\Delta t^4 q}{8} & 0 & \frac{\Delta t^3 q}{3} & 0 & \frac{\Delta t^2 q}{2} \\ \frac{\Delta t^3 q}{6} & 0 & \frac{\Delta t^2 q}{2} & 0 & \Delta t q & 0 \\ 0 & \frac{\Delta t^3 q}{6} & 0 & \frac{\Delta t^2 q}{2} & 0 & \Delta t q \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$